

Figure 7: Run@Local and Run@Cloud

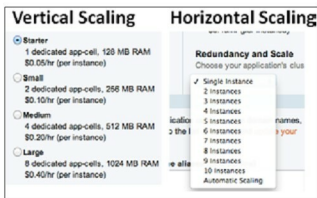


Figure 8: Vertical and horizontal scaling in CloudBees

in adding or removing servers. Tools like New Relic can provide an insight to help you identify what the triggers should be for your application.

Sauce Labs

Sauce Labs is the company behind the very popular Selenium open source project. Selenium is a tool for automated in-browser testing of Web applications. You can develop a Selenium script that operates the website step-by-step within a browser you specify. After a Jenkins server builds the application, you can use Maven to run automated unit tests, and then run Selenium scripts to test the application from the user's perspective, ensuring the build is valid. The entire process can be automated within CloudBees as your application is continuously built and integrated.

Sauce OnDemand is a cloud-based service that allows you to run automated cross-browser functional tests at high speeds, in parallel, so you don't need to maintain testing infrastructure but still get the quality outcome with the cost benefits. Easy-to-use tools let you build both manual and automated tests that produce media-rich results, including screen shots and videos of bugs that can be added to any bug-tracking system.

SonarSource

Sonar is an all-in-one platform to manage source-code quality that provides support to fight the seven mortal sins of the developer: duplicated code, lack of coverage, complexity, no documentation, no standards, potential bugs and a spaghetti design. Sonar offers numerous services to uncover these sins: customisable dashboards, portfolio management, quality profile management, clouds, hotspots and the ability to drill-down to the source code. Sonar is multi-language and integrates well with Eclipse.

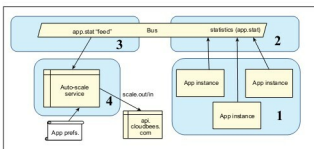


Figure 9: How scaling works in CloudBees



Figure 10: Scaling configuration in CloudBees

The CloudBees SDK

The CloudBees SDK provides command-line tools for your workstation that help to make the use of the CloudBees platform fast and light. Install it so you can build and test your application from your local workstation. Java 6 or higher must be available in the *PATH*.

Download the CloudBees SDK and unzip it into a directory on your machine. Set the *BEES_HOME* environment variable and the *PATH* variable. On first use, enter your CloudBees email address and password.

The SDK includes a set of commands to manage the Run@Cloud Application Service, and can be used to perform most operations that you can do via the Web console. For example, to create a new MySQL database, you'd run the *bees db* command, whose usage is:

```
bees db:create [options] DATABASE_NAME
-p, --password <password> Database password
-u, --username <username> Database username (must be unique)
```



References

- CloudBees White-papers: <http://www.cloudbees.com/resources/white-papers.cb>
- CloudBees—How it Works: <http://www.cloudbees.com/platform/how-it-works.cb>
- CloudBees Documentation:
 - [1] <http://wiki.cloudbees.com/bin/view/Main/CloudBees+Platform+End+to+End+Tutorial>
 - [2] <http://wiki.cloudbees.com/bin/view/DEV/Sauce+OnDemand+Service>
 - [3] <http://wiki.cloudbees.com/bin/view/RUN/Database+Guide>

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